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Research Profile

My research focuses on sepsis, with the goal of connecting patient-level biological and clinical findings to population-level patterns of disease burden. My current work uses transcriptomic and critical-care datasets, computational biology, statistical analysis, multi-cohort harmonization, and mortality modeling. I share version-controlled code, documented pipelines, provenance records, validation outputs, and reusable resources where data-use requirements permit. My research direction is to study demographic and geographic variation, temporal trends, and sepsis burden while developing the epidemiological and health-metrics methods needed for these analyses.

Selected Research & Publications

ClinicalTensorSepsis – a harmonized multi-cohort temporal resource for Sepsis-3 research *Submitted, 2026*
Scientific Data – Data Descriptor

Sole Author & Corresponding Author

🐙 GitHub repository — Cohort construction, temporal reconstruction, cross-database transport

- Prepared a temporal Sepsis-3 dataset using MIMIC-IV v3.1, MIMIC-III v1.4 CareVue, and eICU-CRD v2.0, covering 28,169 adult ICU stays.
- Organized 50 clinical variables into 24-hour trajectories aligned to sepsis onset, using source-specific cohort extraction, measurement mapping, source tracking, and shared data formats.
- Used source-specific methods to reconstruct missing data and organize temporal measurements across databases for external validation and analysis of differences between datasets.
- Released processing code, cohort-flow records, data dictionaries, provenance metadata, and validation outputs through a version-controlled public repository.

36-Gene Blood Transcriptomic Signature Captures Intrinsic Mortality Risk in Early Sepsis *Published, 2026*
Artificial Intelligence in Emergency Medicine, 4 (2026), 100035

First Author & Corresponding Author

DOI: 10.1016/j.aiemed.2026.100035

🐙 GitHub repository — Transcriptomic harmonization, feature selection, external validation

- Created a workflow to analyze sepsis mortality across cohorts using publicly available blood transcriptomic datasets.
- Carried out transcriptomic harmonization, feature selection, statistical analysis, model calibration, and external validation.
- Identified and assessed a 36-gene signature associated with mortality, using SHAP to examine model predictions and functional enrichment to explore biological relevance.

SepsisTensor v1 – A Harmonized Multi-Cohort Transcriptomic Resource for Mortality Prediction *Published May 2026*

Zenodo, Version 1.0

DOI: 10.5281/zenodo.20326923

- Combined seven public sepsis transcriptomic cohorts covering 1,636 patients across microarray and RNA-seq platforms.
- Prepared a shared 7,964-gene dataset using gene mapping, cohort-wise standardization, and ComBat harmonization.

Computational Study of Structural and Functional Effects of EGFR nsSNPs and ncSNPs in Glioblastoma *Under Review, 2026*

Computers in Biology and Medicine

First Author

🐙 GitHub repository — Variant prioritization, structural modeling, regulatory genomics

- Examined coding and non-coding EGFR variants using pathogenicity prediction, structural modeling, regulatory genomics, molecular docking, and molecular dynamics.
- Studied mutation-associated changes in protein stability, molecular interactions, chromatin accessibility, gene regulation, and EGFR–Erlotinib binding.

Current Research

Cross-Database Sepsis Prediction, Phenotyping, and Risk Stratification

Ongoing

- Working on patient-level sepsis models using ClinicalTensorSepsis, including mortality prediction, calibration, external validation, trajectory analysis, phenotyping, and risk stratification.
- Studying how differences between cohorts, measurement patterns, missing data, and database-specific distributions affect model performance across datasets.
- Interested in extending this work to population-level epidemiology, demographic variation, temporal trends, and sepsis burden estimation.

Research Experience

Institute of Bioinformatics and Applied Biotechnology (IBAB), Bengaluru

May – Jun 2026

Research Intern

Project: Molecular dynamics study of mutation-related structural changes in eye lens proteins

Supervisor: Prof. Jayashree Nagesh

Education

Chandigarh University

Punjab, India

Bachelor of Engineering in Biotechnology

Aug 2024 – Jul 2028

CGPA: 8.24 / 10

Technical Skills

Clinical Data

MIMIC-IV, MIMIC-III, eICU-CRD; Sepsis-3 cohort construction, temporal alignment, measurement harmonization, provenance tracking, cross-database distribution shift

Statistics & Validation

Hypothesis testing, bootstrap inference, calibration analysis, external validation, missing-data analysis, PCA, PHATE

Machine Learning

Temporal representation learning, SAITS, DTW, optimal transport, SHAP; scikit-learn, XGBoost, PyTorch

Computational Biology

Transcriptomics, GEO, gene-expression harmonization, differential-expression analysis, functional enrichment, Biopython

Scientific Visualization & Figure Design

R/ggplot2, Matplotlib, ChimeraX, Figma, draw.io; statistical plots, multi-panel figures, scientific schematics

Programming & Research Tools

Python, Bash, Pandas, NumPy, Jupyter, Docker, Git/GitHub

Molecular Modeling

Molecular dynamics, GROMACS, AutoDock Vina, HADDOCK3, PyMOL, ChimeraX

Certification

NPTEL – Structural Biology

Jan – Apr 2026

Elite + Silver